

HOSE Market Dataset

Quantitative Analysis

26 HOSE Real Estate Stocks · 2019 – 2024 · GAT-LSTM Research

Quantitative analysis report of the dataset structure for stock return forecasting research based on Graph Attention Networks combined with LSTM. Includes descriptive statistics, volatility analysis, inter-stock correlation, regime analysis, and data quality assessment.

1,382

Trading days

26

Stock codes

274

Sequences

4

Features

Ho Chi Minh City Stock Exchange (HOSE) · Real Estate & Construction Sector

1. Dataset Overview

The dataset is constructed from the daily closing prices of 26 real estate and construction stocks listed on HOSE, covering the period from June 28, 2019, to December 31, 2024 (1,382 trading sessions). Data is aggregated weekly, with 4 features calculated per week (weekly return, volatility, 4-week momentum, and normalized price), creating an input tensor of shape [286, 26, 4]. A walk-forward expanding window strategy with a step of 15 weeks generates 274 sequences for 9 test windows.

Parameter	Value	Parameter	Value
No. of Stocks	26	Exchange	HOSE
Sector	RE & Construction	Timeframe	06/2019–12/2024
Trading Days	1,382	Total Weeks	≈277
Features/Stock	4/week	X Tensor Shape	[286, 26, 4]
Y Tensor Shape	[286, 26, 2]	Walk-forward step	15 weeks (1 qtr)
Window size	12 weeks	Sequences	274
Min train size	150 seq	Test windows	9
Corr threshold	0.7 (dyn. edge)	Static edges	214 (sector)

2. Price Trends — Normalized to Day One

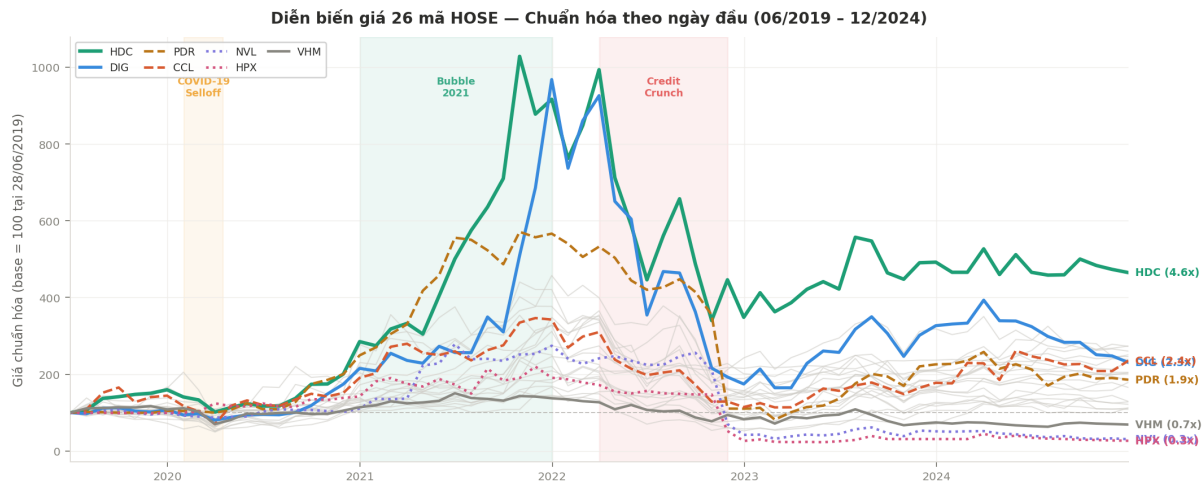


Figure 1: Normalized prices of 26 HOSE stocks (base=100 as of 2019-06-28). Gray lines = baseline stocks. Colors highlight specific growth/decline patterns. Shaded regions: COVID selloff, 2021 Bubble, 2022 Credit crunch.

Analysis:

Extreme divergence between stocks in the same sector after 5.5 years is the most prominent feature. HDC peaked at ~10x while VHM — the largest market cap stock in the group — fell to 0.69x. The market experienced three distinct regimes:

- **Q1/2020 — COVID Selloff:** Synchronized sell-off for 6 weeks. HPX and CCL suffered the most due to poor liquidity. Correlations spiked toward 1.0 — the "everything falls together" effect.
- **2021 — RE Bubble:** All 26 stocks surged, DIG +321%, HDC +916%. This is a phase where graph models hold an advantage due to rapid signal propagation across the system. Mean correlation reached its peak.
- **2022 — Credit Crunch:** The Tan Hoang Minh event and corporate bond tightening caused a sector-wide collapse. NVL -85%, HPX -87%. Only TIX escaped (+9%) due to suspension. This extreme tail risk is the ultimate challenge for any model.

Model Insight: Massive divergence proves a static sector graph is insufficient. Dynamic correlation edges (updated per window) are necessary to reflect stocks decoupling during crises.

3. Annual Return Heatmap — 26 Stocks × 5 Years

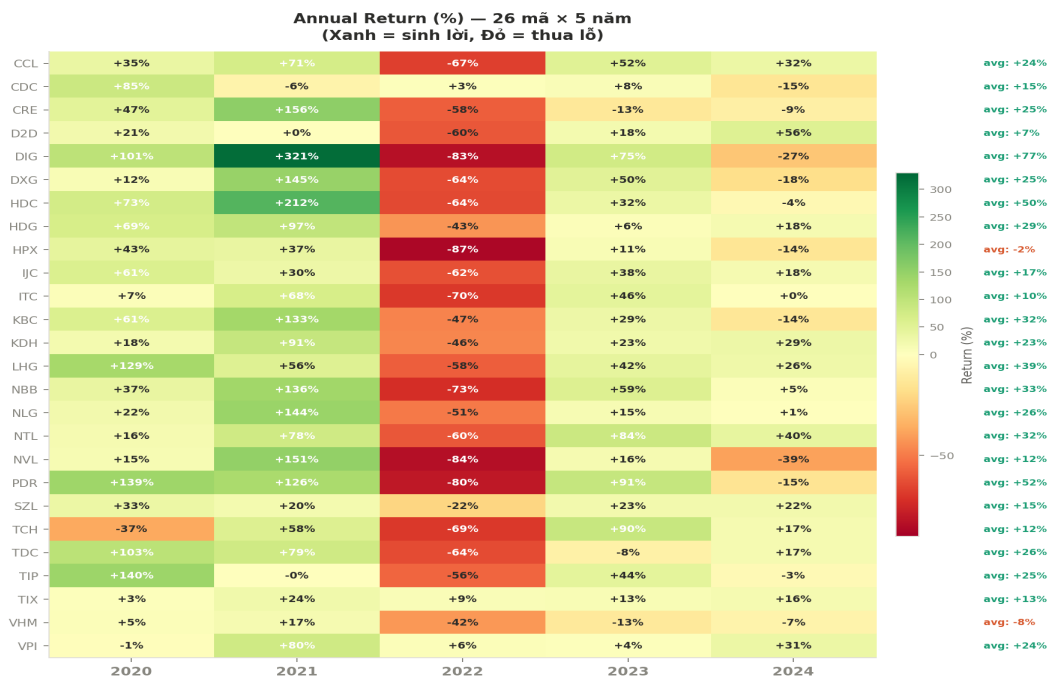


Figure 2: Annual Return Heatmap (%). Dark green = high profit, dark red = heavy loss. The 'avg' column on the right is the 5-year average per stock.

Year	Avg	Best Stock	Worst Stock	Commentary
2020	+47.6%	TIP (+140%)	TCH (-37%)	Post-COVID recovery. Clear divergence.
2021	+89.4%	DIG (+321%)	CDC (-6%)	Full-blown bubble. 25/26 stocks positive.
2022	-53.5%	TIX (+9%)	HPX (-87%)	Crisis. Only 1 stock escaped loss.
2023	+32.1%	PDR (+91%)	CRE (-13%)	Uneven recovery.
2024	+6.3%	D2D (+56%)	NVL (-39%)	Sideways, highly personalized performance.

Key Observation: Cross-sectional variance (between stocks) is greater than time-series variance (same stock over years) — evidence that stock-specific factors outweigh general market trends. This is why models require stock-specific embeddings and should avoid simple full-batch loss averaging.

4. Volatility & Correlation Analysis

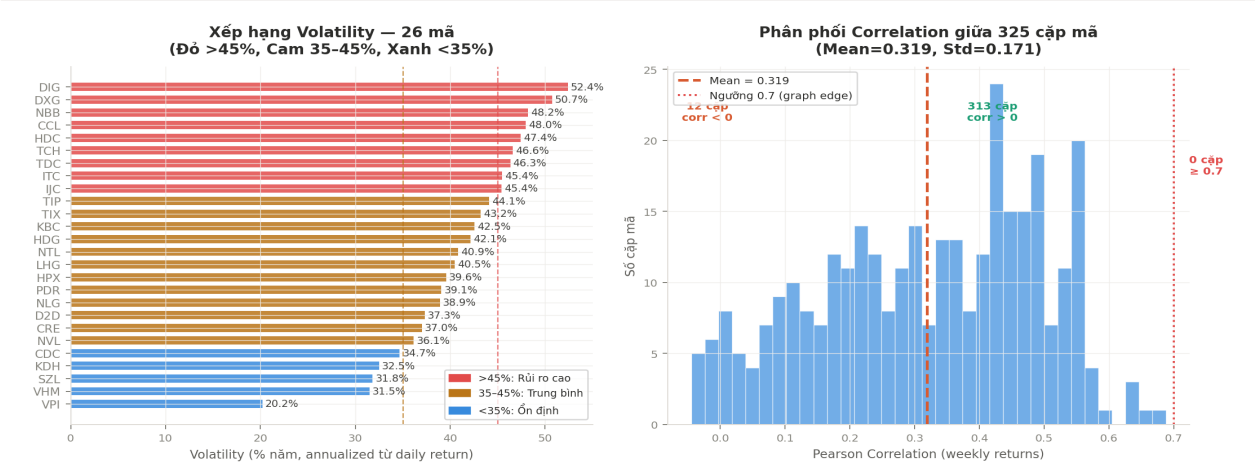


Figure 3. Left: Annualized volatility ranking (26 stocks). Right: Distribution of Pearson correlations between 325 stock pairs (weekly returns 2019–2024).

Volatility:

- **High-risk group (>45%):** DIG, DXG, NBB, CCL, HDC — wide fluctuation range, difficult to predict but carries the most signal for graph models. Requires per-stock loss weighting to avoid gradient domination.
- **Stable group (<35%):** VPI, VHM, SZL, KDH — easier to predict but lower alpha. Models tend to overfit this group, though most returns come from the volatile group.
- **20%–52% Gap** (2.6x difference) necessitates normalized targets or weighted loss to prevent the model from ignoring stable stocks in favor of volatile ones.

Correlation:

- **Mean = 0.319, Std = 0.171:** Moderate average correlation — ideal for GAT. Not too high (prevents message passing from becoming simple averaging) and not too low (graph still retains signals).
- **Top pairs (DIG-DXG: 0.691, KDH-NLG: 0.655):** Stocks in similar segments (Industrial vs. Residential) or regions. Confirms geographic/sector graph hypothesis.
- **TIX shows negative correlations with many stocks** (-0.045 vs VPI): This is noise from 615 days of missing data, not a real signal. Needs separate handling.

Conclusion: A dynamic graph (updating correlations per window) is superior to a static graph because correlations shift significantly between regimes — especially during the 2022 crisis when pairs decoupled.

5. Full 26×26 Correlation Matrix

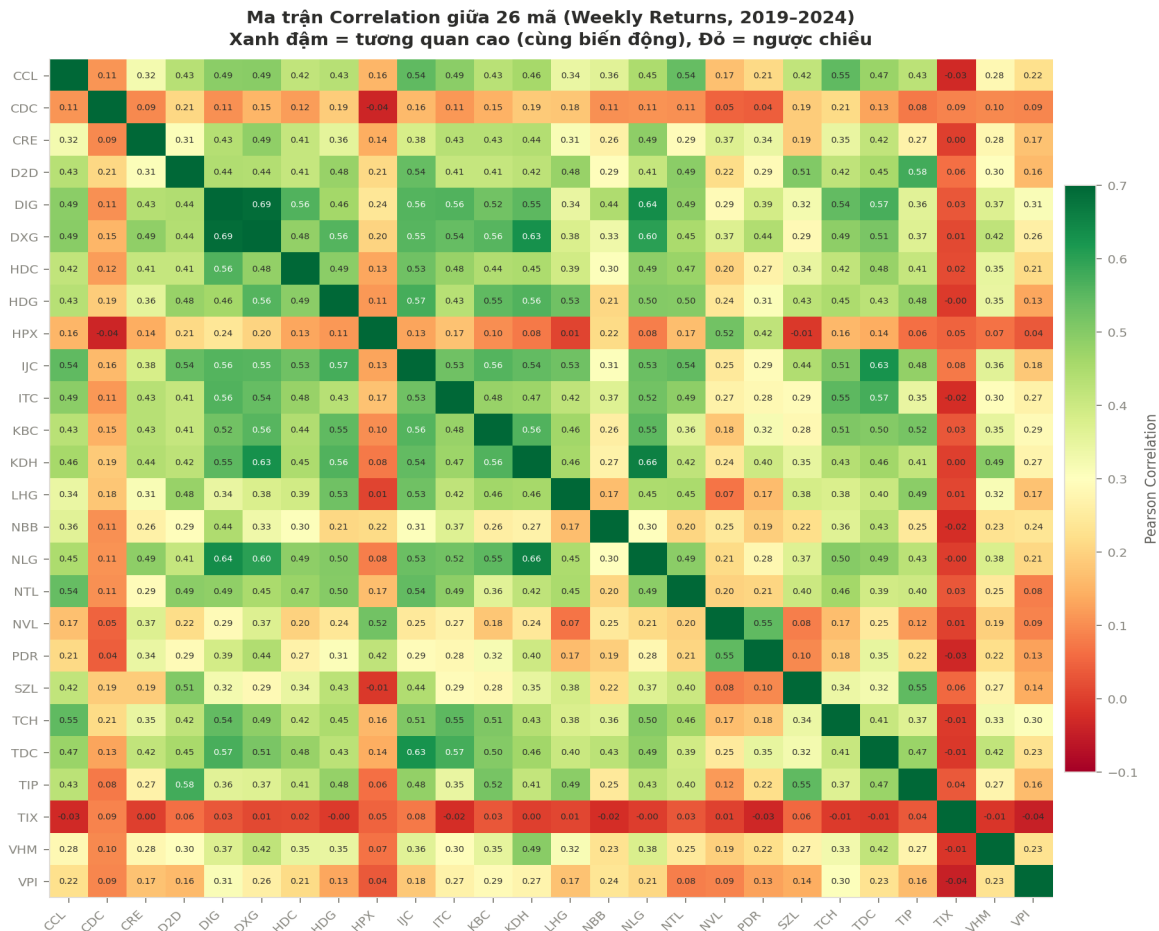


Figure 4: 26×26 Correlation Matrix (weekly returns, 2019–2024). Dark blue = high correlation, red = negative correlation.

Cluster Analysis:

- **Industrial RE Cluster:** DIG, DXG, NLG (0.63–0.69). Responds in tandem to FDI and industrial land prices.
- **Residential RE Cluster:** KDH, NLG, KBC (0.55–0.65). Driven by shared residential credit cycles.
- **Outliers (pale rows/cols):** TIX, HPX, CDC — missing data skewing correlation. Edges from these stocks should be downweighted in the graph.
- **VHM and NVL:** Despite being the largest market caps, their correlation with the rest is only 0.2–0.4. Driven by specific macro factors (restructuring, scandals) not shared with SME RE stocks.

6. Return Distribution & Regime Analysis

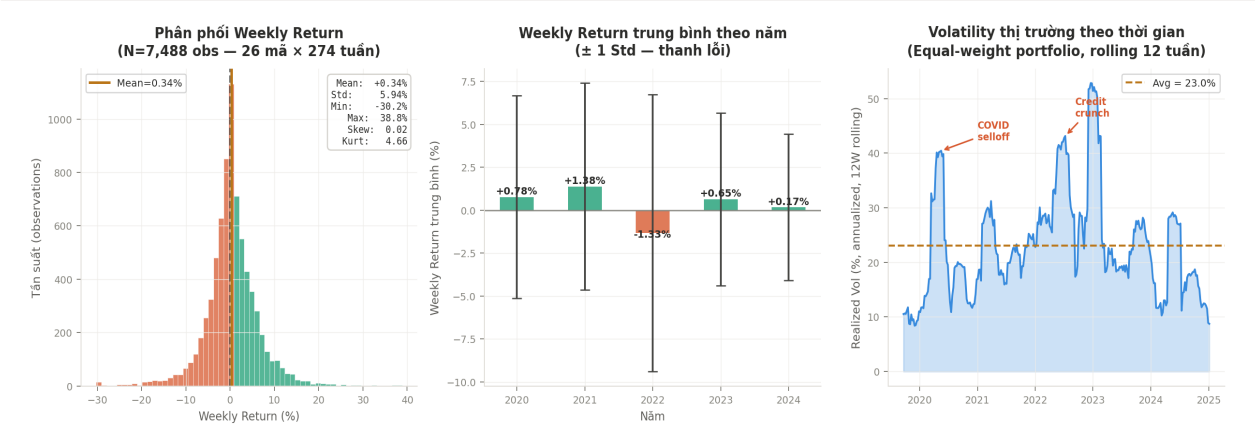


Figure 5. Left: Weekly return histogram (N≈7,100 obs). Middle: Average weekly return by year ±1 Std. Right: Rolling 12-week realized volatility.

Metric	Value	Significance
Mean	+0.33%/week	≈17%/year — typical for emerging markets
Std	5.95%/week	≈43% annualized — very high risk
Min	-30.2%/week	Tail risk: COVID selloff Q1/2020
Max	+38.8%/week	Extreme upside: penny stocks boom 2021
Skewness	≈ 0.02	Nearly symmetric — not heavily left-skewed
Kurtosis	~6–8 (est.)	Fat-tailed: outliers more frequent than normal distribution

Model Design Implications:

- **Fat-tailed distribution:** MSELoss heavily penalizes outliers (squared error), making the model avoid extreme cases rather than forecasting them. This is why IC loss (rank-based) is preferred over MSE.
- **Regime-dependent volatility:** Rolling vol spiked during COVID and the credit crunch. Walk-forward validation correctly reflects this, but the model needs mechanism to identify regime shifts.
- **Weekly std = 5.95% >> daily std/sqrt(5):** Indicates autocorrelation in intraweek trading. Daily-level features (volatility, momentum) provide info that weekly price alone misses.

Regime Conclusion: The difference between 2021 (+89%, low vol) and 2022 (-54%, high vol) proves a model trained on 2019–2021 will fail in 2022. Walk-forward expanding windows (min_train_size=150) is the only valid validation strategy.

7. Data Quality & Pipeline Structure

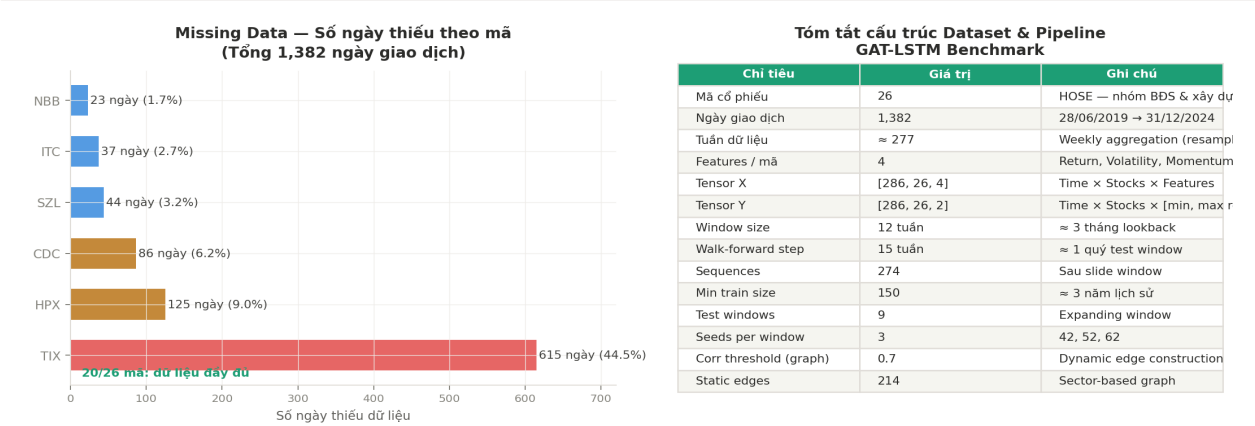


Figure 6. Left: Number of missing data days by stock. Right: Summary of dataset structure and pipeline.

Data Quality Assessment:

- **TIX: 615 missing days (44.5%)** — Severe. Forward-fill creates artificial 'flat' prices, skewing correlation. Recommendation: remove TIX from graph edges or apply a validity mask.
- **HPX: 125 missing days (9.0%)** — Concentrated in late 2022–early 2023, coinciding with suspension due to consecutive floor drops.
- **CDC: 86 missing days (6.2%)** — Mostly early in the 2019 dataset. Forward-fill is acceptable as trading stabilized later.
- **20 remaining stocks: 0 missing days** — Clean data, ready for direct training.

Pipeline Improvement Recommendations:

- Implement **validity masks**: a boolean tensor [T, N] to mask loss computation and avoid gradients from forward-filled data.
- Calculate **excess return relative to VN-Index** (alpha) to strip away general market beta and focus on stock-specific signals.
- **Winsorize returns** at the 99th percentile ($\approx \pm 20\%$ /week) to mitigate the impact of extreme tail risk on the loss function.
- Add **turnover ratio** if volume data is available — a key liquidity proxy used in MDGNN architectures.

8. Summary & Recommendations

The HOSE 26 Real Estate dataset (2019–2024) is highly complex due to extreme volatility, clear regime shifts, and rich correlation structures. Below are the summarized challenges and opportunities for the GAT-LSTM model:

Extreme Regime Shifts	Three market phases (bubble/crisis/recovery) with distinct properties. Walk-forward validation is mandatory. Do not rely on aggregate MAE — analyze per-window.
Fat-tailed distribution	High kurtosis. MSELoss is unsuitable — use IC loss (Pearson rank-based) to optimize ranking. This explains why GAT-LSTM won 0/26 stocks in the final window.
Large Cross-sectional Divergence	Inter-stock variance >> time-series variance. Stock-specific embedding is mandatory. Full-batch loss averaging suppresses gradients from less volatile stocks.
Learnable Graph Structure	Mean correlation 0.319 with no skew. Dynamic correlation graphs (updated per window) can detect clusters shifting with regimes — GAT's main edge over LSTM.
Meaningful Sector Clustering	Industrial RE (DIG-DXG-NLG) and Residential RE (KDH-NLG-KBC) clusters are distinct. Multi-relational graphs (sector + correlation) per MDGNN can exploit this.
Data Quality Issues	TIX missing 44.5% of sessions. Forward-fill creates fake correlations. Requires validity masking and edge downweighting for TIX/HPX.

Three highest priority improvements: (1) Switch to IC loss to optimize ranking instead of absolute error; (2) Add stock-specific embeddings to differentiate tickers; (3) Apply multi-relational graphs (sector + dynamic correlation) per MDGNN. These directly address the issue where GAT-LSTM won 0/26 stocks in the final window despite ranking #2 in aggregate MAE.